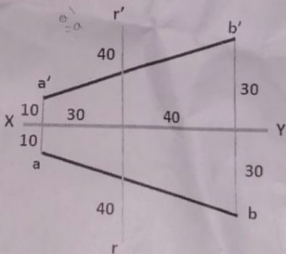


Department of Computer Science & Engineering, Punjabi University, Patiala
Engineering Graphics, First Mid-Semester Exam, September, 2025

B. Tech. Ist Year, 1st Sem (CSE5678)

Max. Marks: 15

NOTE: Section A is compulsory. Attempt any two questions from Section B.

Q.1	Section A: Answer the following five questions with pen only.	
(i)	The point P lies in 2 nd quadrant, its front view will lie _____ XY reference line and top view will lie _____ XY reference line. (above / below / on)	01
(ii)	Give the dimensions of A4 size drawing sheet.	01
(iii)	The true length of a line will appear on a reference plane, when the line is _____ to the RP.	01
(iv)	An AIP is _____ to HP and _____ to VP. (parallel / perpendicular / inclined)	01
(v)	If a pentagonal lamina is inclined to VP and perpendicular to HP, its projection on HP will reveal its _____ view. (true / apparent / edge)	01
Section B: Draw any two out of the following questions.		
Q. 2	A point 35 mm above XY line is the plan-view of two points P and Q. The elevation of P is 45 mm above the H.P. while that of the point Q is 40 mm below the H.P. Draw the projections of the points and state their position with reference to the principal planes and the quadrant in which they lie.	05
Q. 3	A line PQ has its end P 40 mm above HP and in V.P. and end Q 45 mm in front of V.P. & 5 mm above H.P. The distance between the end projectors of the line is 50 mm. Draw the projections of the line & find its true length (T.L.), inclinations θ & Φ . Locate the H.T. & V.T. of the line using rotation of line method.	05
Q. 4	Find the shortest distance of the line connecting point R to the line AB as shown in figure 1. Show the line of the shortest distance in plan and elevation. 	05
5	A regular pentagonal lamina of 25 mm edge, has one of its corners on H.P. The plane of the lamina is inclined at 45° to the H.P. The side of the pentagon, which is opposite to the corner (which is on H.P.) is inclined at 30° to the V.P. Draw the projections of the lamina.	05

Department of Mechanical Engineering, Punjabi University, Patiala
MCE -151 Engineering Graphics
Second Mid-Semester Exam, May-2022
B. Tech. 1st Year, 2nd Sem (Group A)

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Time: 1 Hour

Max. Marks: 15

NOTE: Section A is compulsory. Attempt any two questions from Section B.

Section A		
Q. 1	<u>All five parts of this question are compulsory</u>	(05)
(i)	A Tetrahedron is solid that falls under the category of _____ (Regular Polyhedra / Prism / Pyramids)	01
(ii)	The development of a cylinder consists of _____. (Isosceles triangles, Sector of a circle / Rectangle / Parallelogram)	01
(iii)	A pentagonal pyramid resting on the ground on its base is cut by a cutting plane parallel to the horizontal plane. The true shape of the cut section will appear in _____ view. (Front / Top / side)	01
(iv)	A cone 40 mm diameter and 60 mm height, resting on the ground on its base is cut at its axis 20 mm below the apex, by an AIP inclined at 30° to HP. The true shape of the cut section will be _____. (Circle / Parabola / Ellipse / Hyperbola)	01
(v)	The isometric projection of a square is _____. (Square, Rectangle, Parallelogram, Rhombus)	01
Section B		
<u>Draw any two out of the following three questions.</u>		
Q. 2	A cube of 30 mm side rests with one of its edges on HP such that one of the square faces containing that edge is inclined at 30° to HP and the edge on which it rests being inclined to 60° to VP. Draw its projections.	(05)
Q. 3	A cylinder of 40 mm diameter, 60 mm height and having its axis vertical is cut by a section plane, perpendicular to the VP, inclined at 45° to the HP and intersecting the axis 32 mm above the base. Draw its front view, sectional top view, and the true shape of the cut section.	(05)
Q. 4	Draw an Isometric Projection of a Sphere of diameter 40 mm resting centrally on top of a rectangular block of base 40 mm x 60 mm and height 30 mm.	(05)

Engineering Graphics, Second Mid-Semester Exam, April, 2025
B. Tech. 1st Year, 1st Sem (CSE5678)

Time: 1 Hour

Max. Marks: 15

NOTE: Section A is compulsory. Attempt any two questions from Section B.

Section A: Answer the following five questions with pen only.		
Q.1		
(i)	The right solid is defined as a solid in which axis is inclined at _____ degree to the base. (30 / 45 / 90 / 120)	01
(ii)	The number of edges in a hexagonal prism are _____. (9 / 12 / 18 / 24)	01
(iii)	The truncated solid is obtained by cutting the solid _____ to its base. (parallel / perpendicular / inclined)	01
(iv)	Isometric projections are bigger than isometric drawings. (True / False)	01
(v)	Radial line development method is used for the development of _____. (Cube / Pentagonal prism / Pyramid)	01
Section B: Draw any two out of the following three questions.		
Q. 2	A right regular Hexagonal pyramid, edge of base 25 mm and height 50 mm, rests on one of its base edges in HP with its axis parallel to VP. Draw the projections of the pyramid when its axis inclined at 30° to HP.	05
Q. 3	A right regular hexagonal pyramid of base side 25 mm, axis 50 mm long is resting with its base on HP and one an edge of the base is perpendicular to VP. It is cut by a section plane perpendicular to VP and inclined to HP at 30° and passing through a point 20 mm below the vertex. Draw the front view, sectional top view and the true shape of the section.	05
Q. 4	A cube of edge 30mm rests centrally on top of a square block of base edge 45mm and height 20mm. Draw the isometric projections of the combined solid.	05